



THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFERABAD

DEPARTMENT OF SOFTWARE ENGINEERING

Computer Networks

(Lab no # 03)

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Submitted To:

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Computer Networking Lab Report

Title: Ethernet Cable Crimping and Testing

1. Objective

The objective of this lab was to learn how to prepare an **Ethernet LAN cable**, attach an **RJ45 connector**, and verify the connection using a **LAN cable tester**. This practical activity helped us understand the physical layer of computer networking and how network cables are made and tested.

2. Tools and Materials

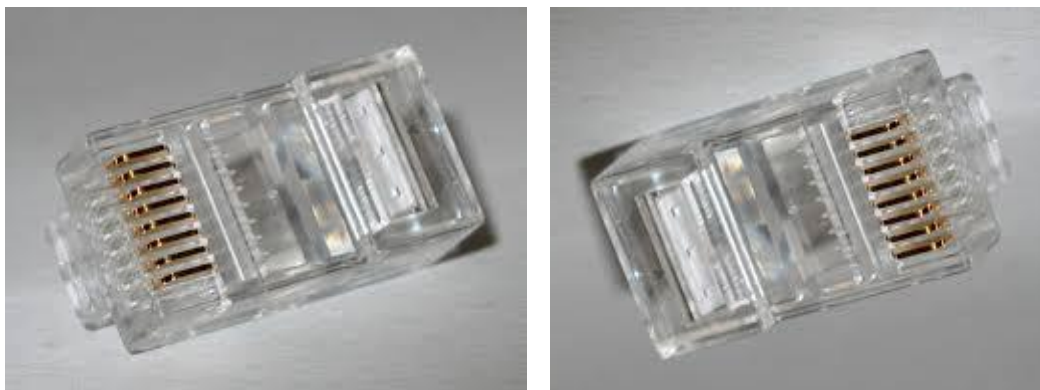
1. Ethernet Cable (Cat5e / Cat6)

Used to transmit data between network devices such as computers, switches, and routers.



2. RJ45 Connector

A connector attached to the end of the Ethernet cable to connect it to network devices.



3. Cable Stripping Tool

Used to remove the outer insulation of the Ethernet cable so the internal wires can be accessed.



4. Crimping Tool

Used to press the RJ45 connector onto the cable so the wires are securely connected to the pins inside the connector.



5. LAN Cable Tester

Used to check whether the Ethernet cable is correctly wired and functioning properly.

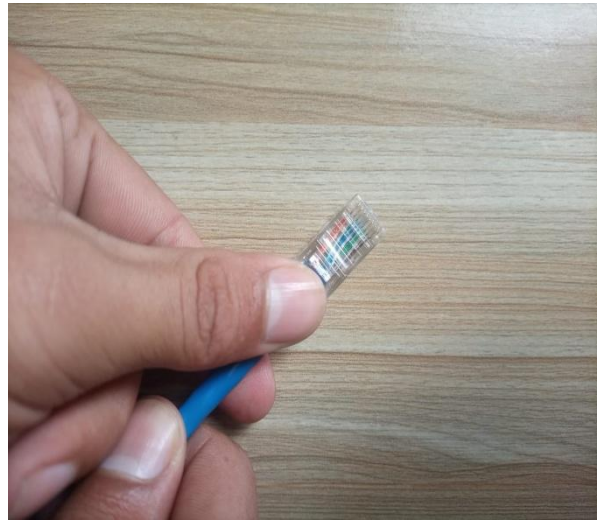
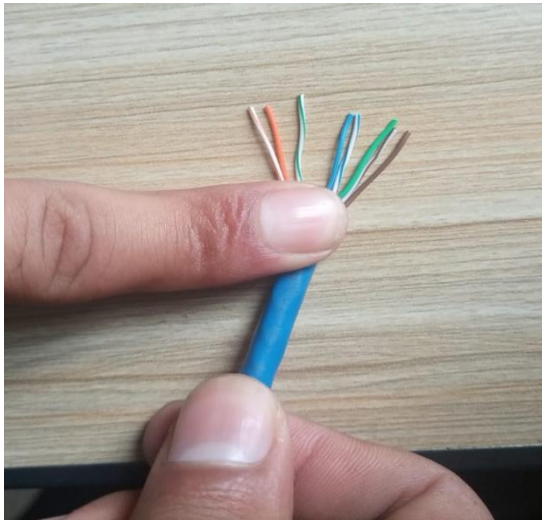


3. Procedure

1. First, an **Ethernet cable** was taken and the outer insulation was removed using a **cable stripping tool**.
2. After removing the cover, the internal wires were separated and straightened carefully.
3. The wires were then arranged according to the **standard color sequence (T568A or T568B)**.
4. After arranging the wires, they were inserted into the **RJ45 connector** in the correct order.
5. The connector was then placed into the **crimping tool** and pressed firmly so that the metal pins inside the connector could attach to the wires.
6. Finally, the cable was connected to a **LAN cable tester** to verify that all wires were correctly connected and the cable was working properly.

4. Result

The Ethernet cable was successfully created and tested. The LAN cable tester confirmed that all wires were properly connected and the cable was functioning correctly.



5. Conclusion

This lab provided practical knowledge about **Ethernet cable preparation and testing**. We learned how to strip the cable, arrange the wires according to standard networking methods, attach an RJ45 connector using a crimping tool, and verify the cable using a LAN tester. This activity helped us understand the importance of correct wiring for reliable network communication.